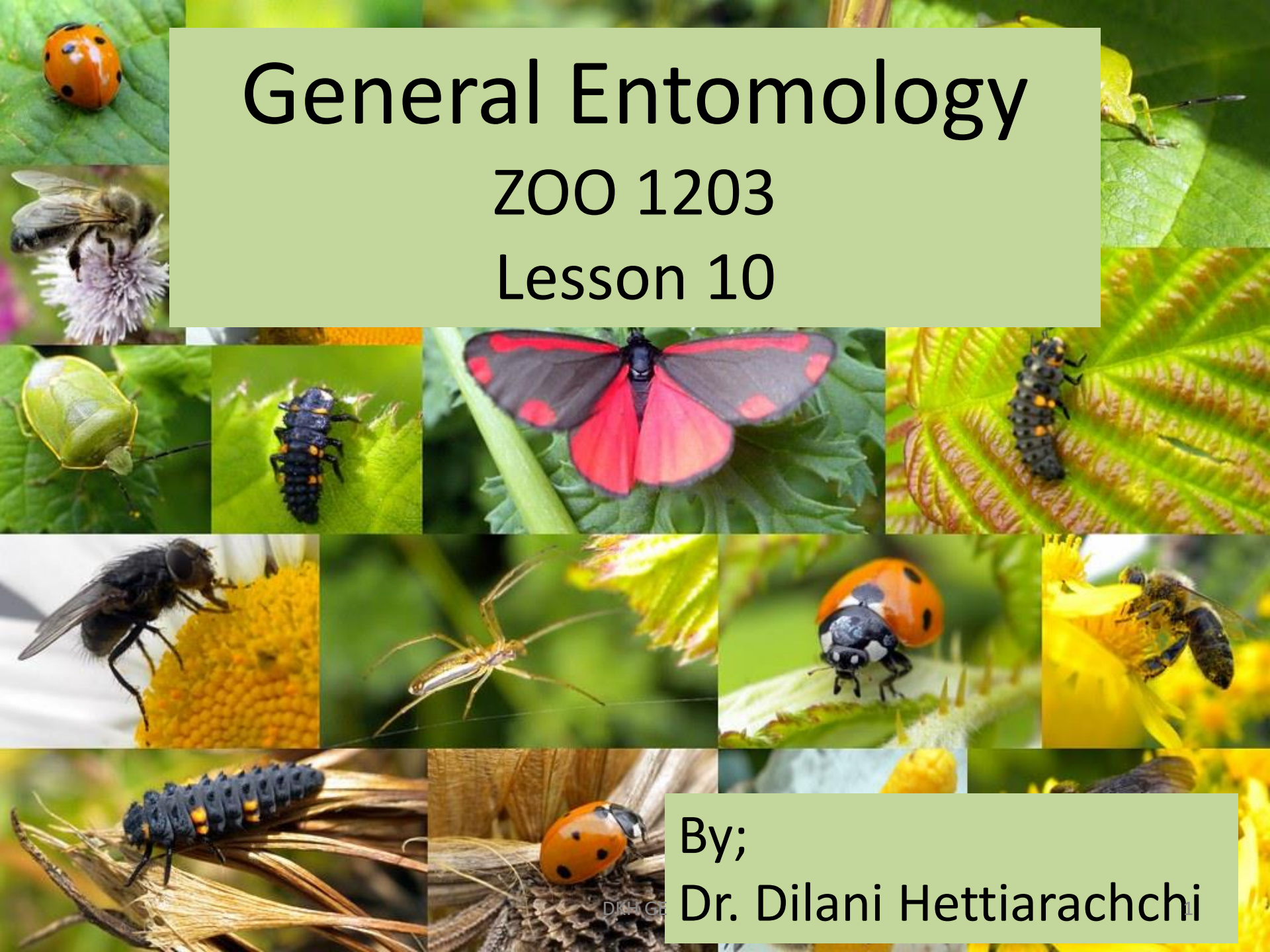


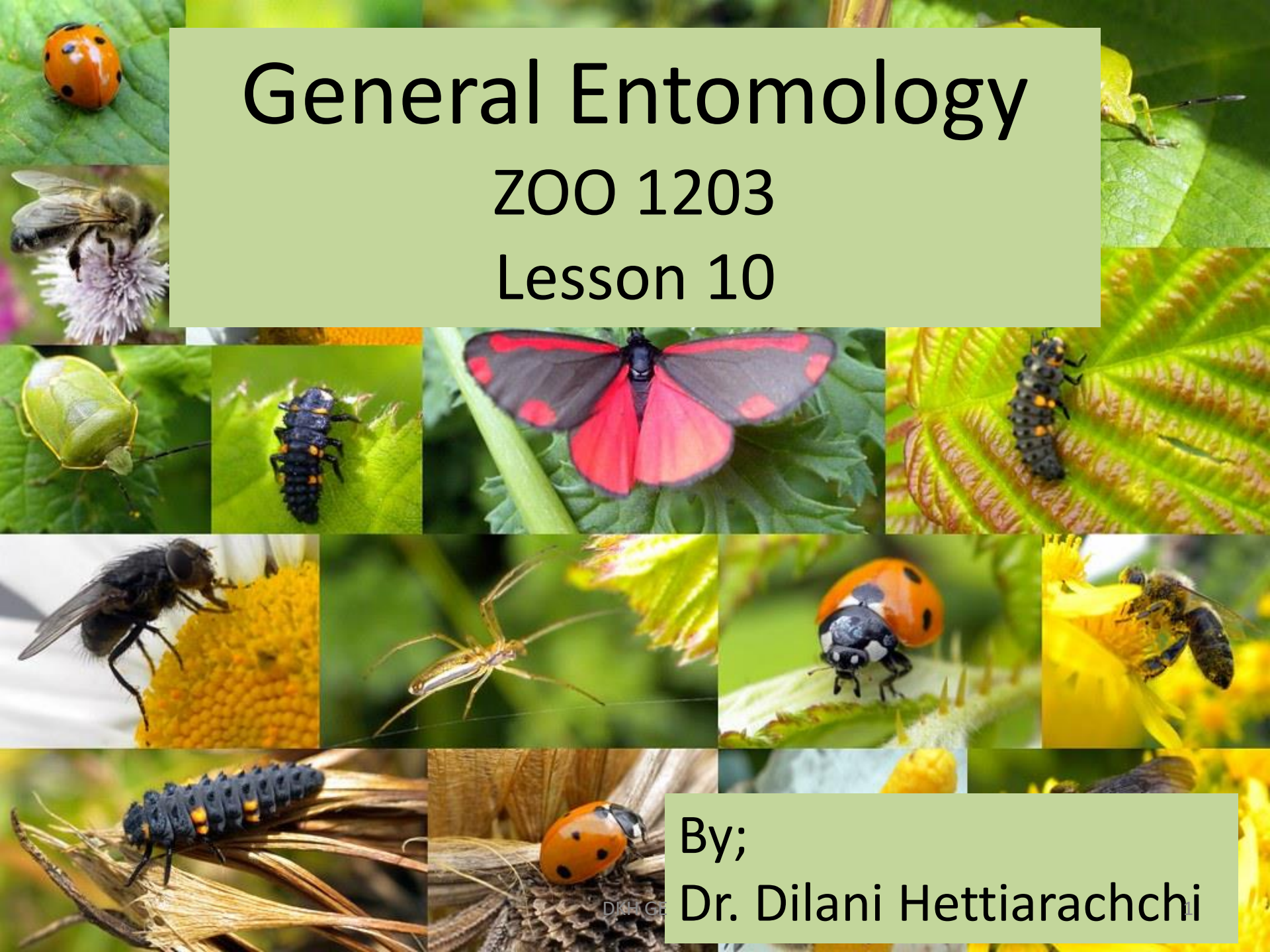
A collage of various insects including ladybugs, bees, flies, and caterpillars on leaves and flowers. The insects are shown in different poses and environments, highlighting their diversity. The background is a light green gradient with a subtle pattern of small white dots.

General Entomology

ZOO 1203

Lesson 10

By;
Dr. Dilani Hettiarachchi

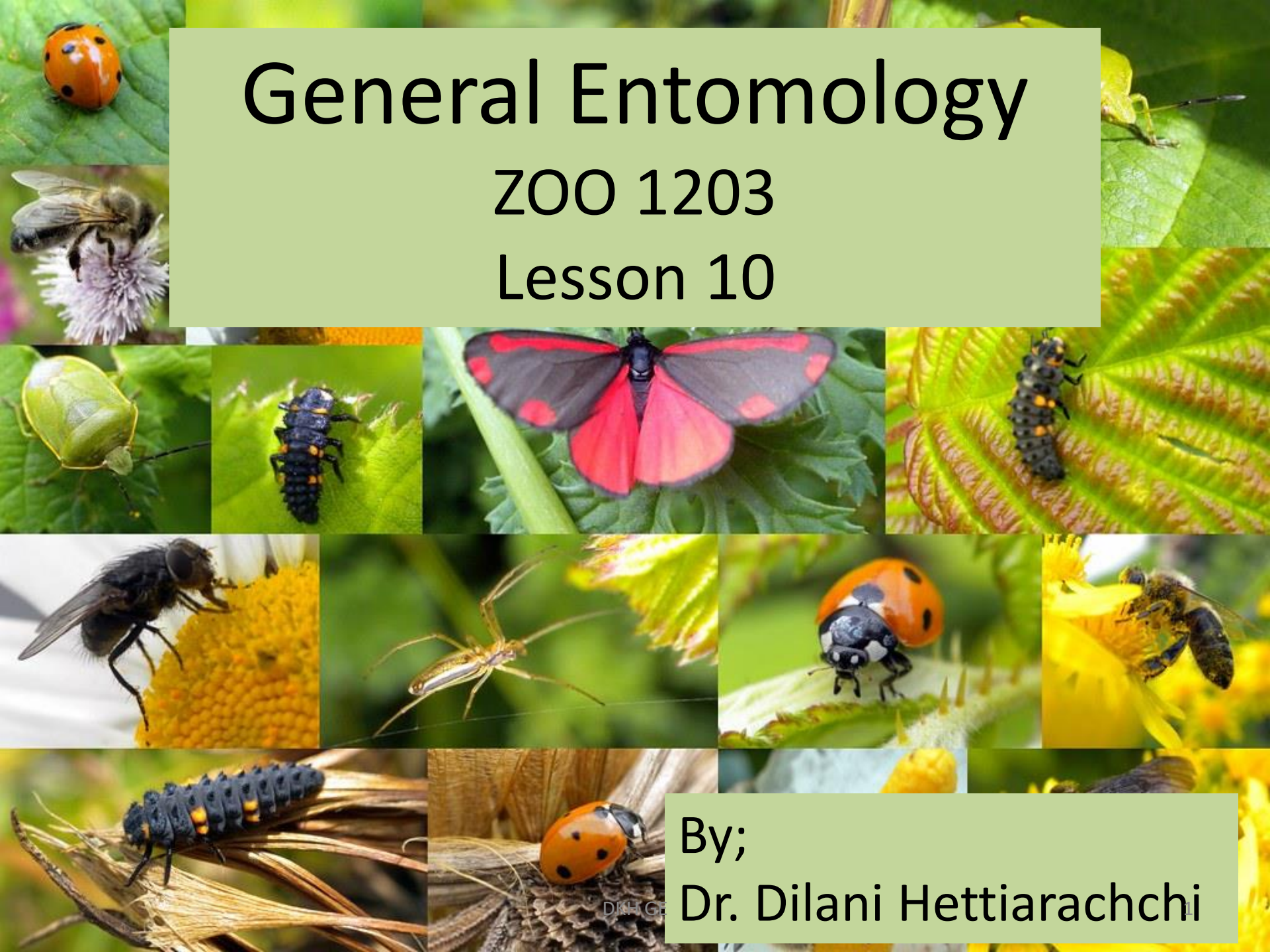
A collage of various insects including ladybugs, bees, flies, and caterpillars on leaves and flowers. The insects are shown in different poses and environments, highlighting their diversity. The background is a light green gradient with a subtle pattern of small white dots.

General Entomology

ZOO 1203

Lesson 10

By;
Dr. Dilani Hettiarachchi

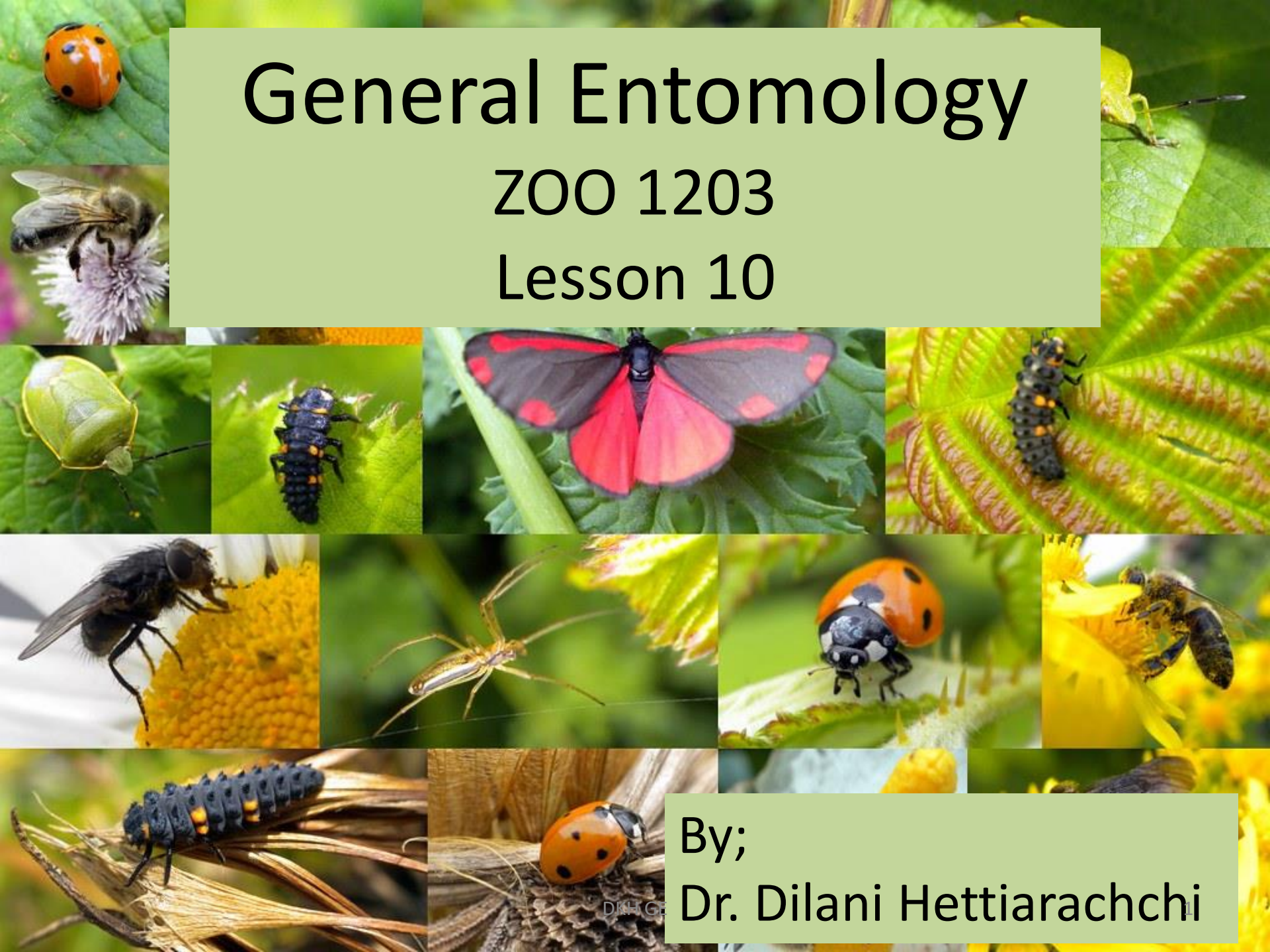
A collage of various insects including ladybugs, bees, flies, and caterpillars on leaves and flowers. The insects are shown in different poses and environments, highlighting their diversity. The background is a light green gradient with a subtle pattern of small white dots.

General Entomology

ZOO 1203

Lesson 10

By;
Dr. Dilani Hettiarachchi

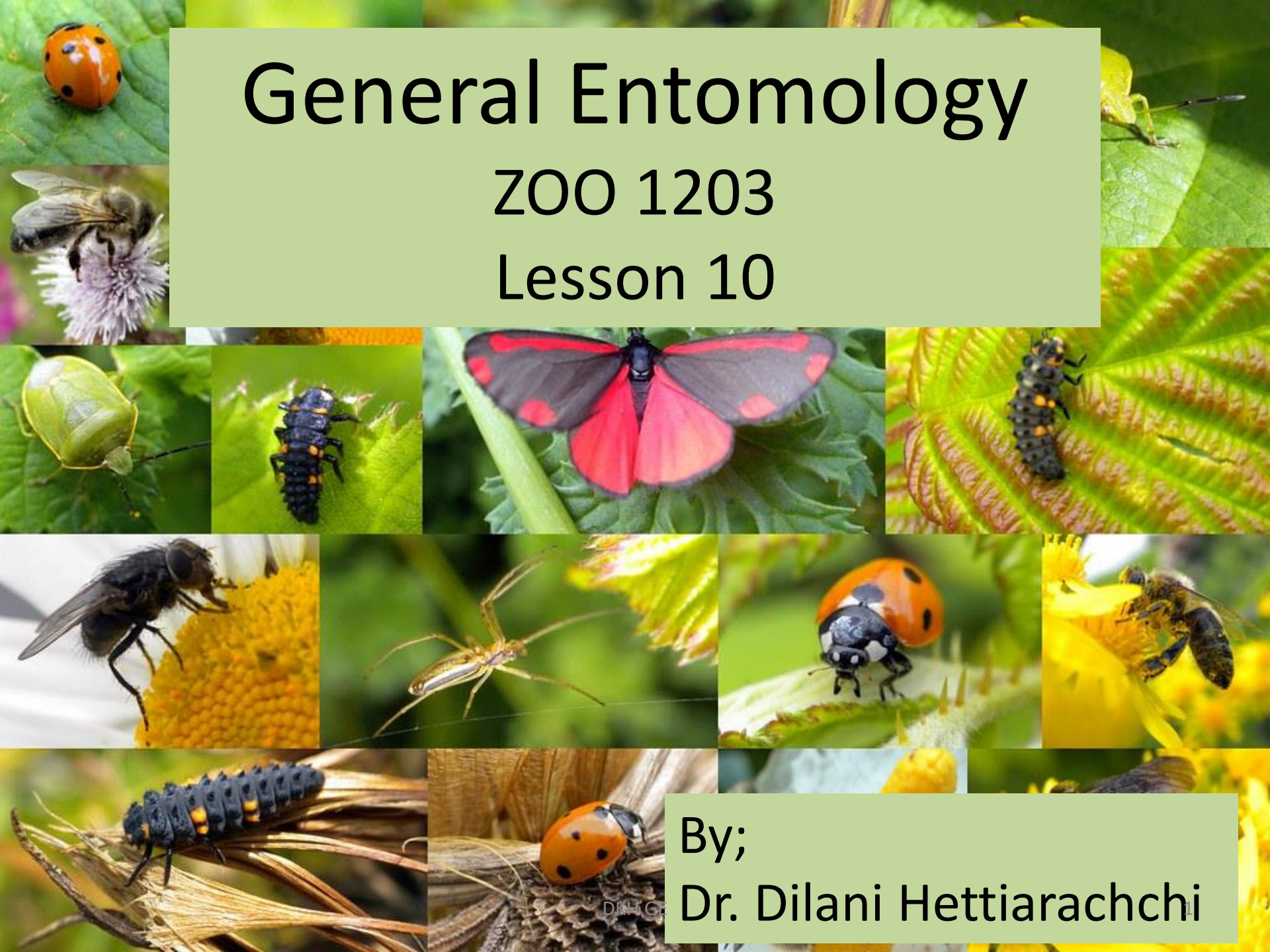
A collage of various insects including ladybugs, bees, flies, and caterpillars on leaves and flowers. The insects are shown in different poses and environments, highlighting their diversity. The background is a light green gradient with a subtle pattern of small white dots.

General Entomology

ZOO 1203

Lesson 10

By;
Dr. Dilani Hettiarachchi

A collage of various insects including ladybugs, bees, flies, and caterpillars on leaves and flowers. The insects are shown in different poses and environments, highlighting their diversity. The background is a light green gradient with a subtle pattern of small white dots.

General Entomology

ZOO 1203

Lesson 10

By;
Dr. Dilani Hettiarachchi

Lesson Objectives

After completion you should be able to;

- Describe special features of major orders of insects
- Identify insects up to order level

Insect Classification

- over one million species have been described to date
- classification of insects can be complex but it is very important to group and identify insects so that they can be studied reliably
- classified using a hierarchical system of classification
- binomial nomenclature

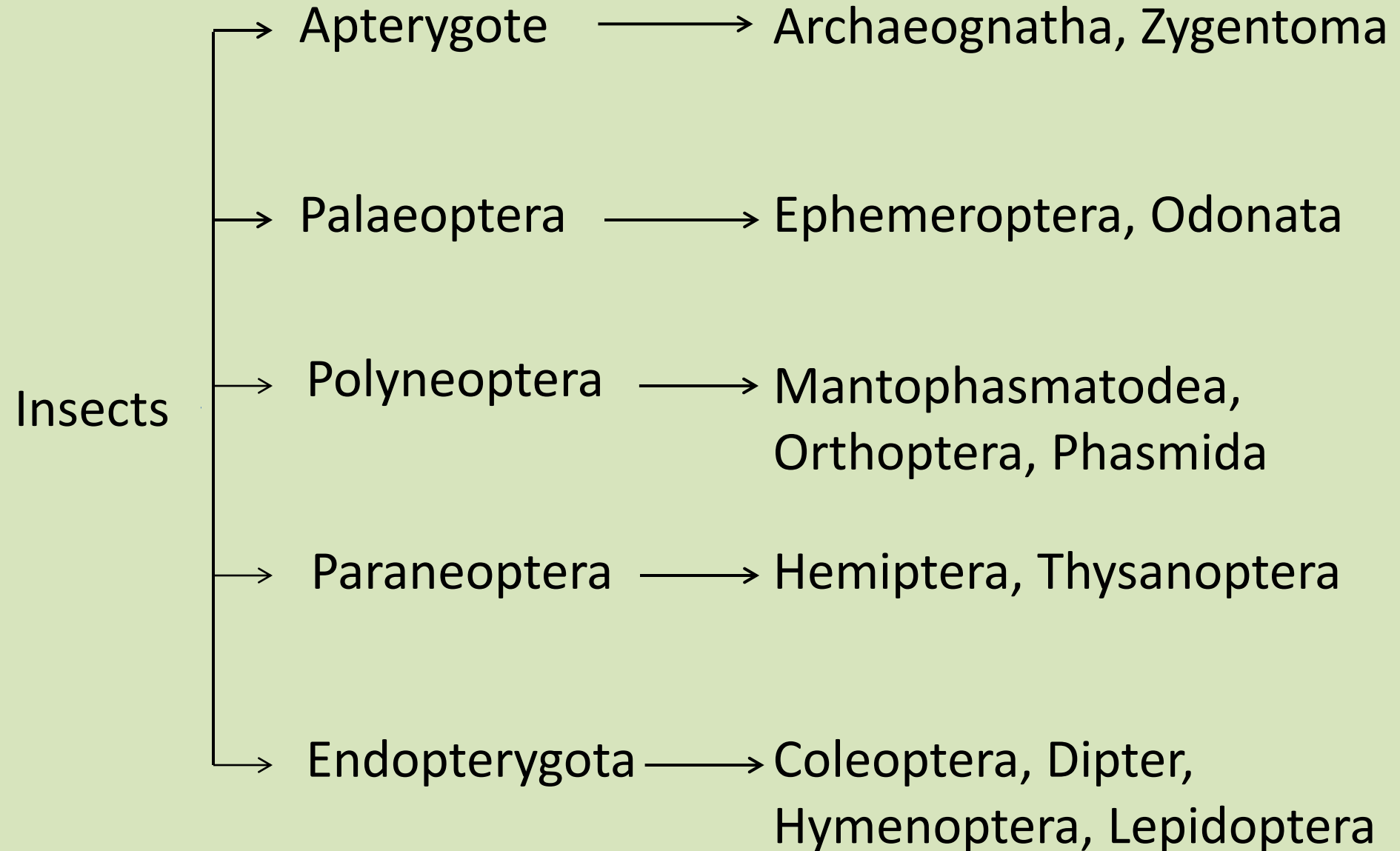
Kingdom: Animalia (all animals)

Phylum: Arthropoda (all arthropods)

Class: Insecta (only the insects)

Sub-class

Some Orders



Sub-Class : Apterygota

- Wingless primitive insects

Order: Archaeognatha/Microcoryphia

Eg: Bristletails

- Meaning “ancient jaw” - primitive (monocondylic)
- Active cylindrical insects
- Up to 18 mm long
- Have a long flagellate antennae
- Abdomen bears a long terminal filament flanked by a pair of cerci
- This gives three tailed appearance
- Mouthparts – notable due to maxillary palp (7 segments): longer than the legs

- Much of the body covered with flat scales
- Gives a shiny appearance
- Ametabolous development

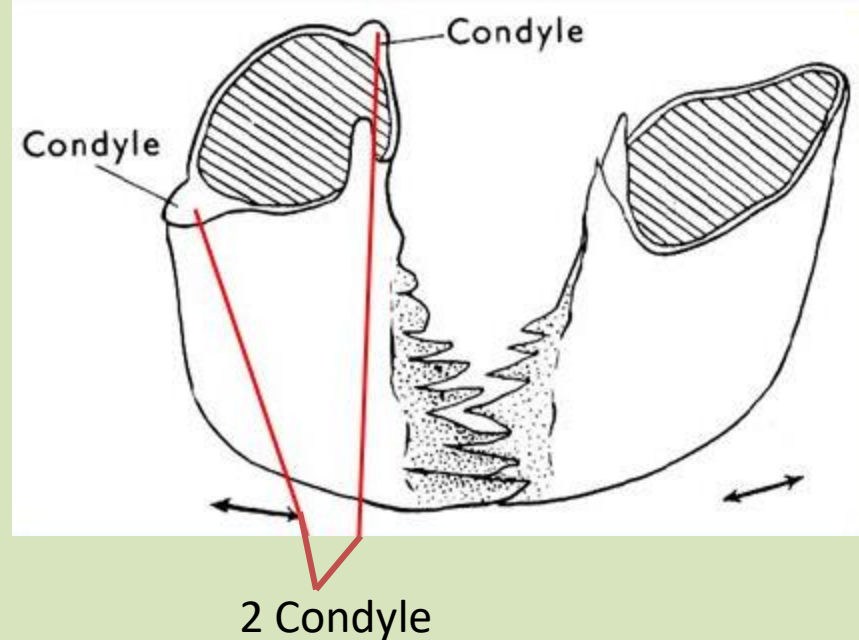
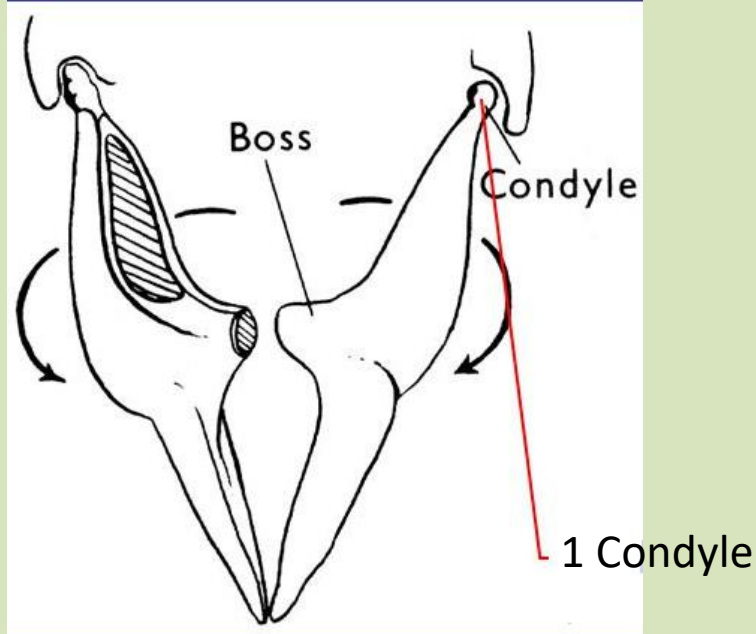


Bristletails are common inhabitants of forest leaf litter. They are part of the community of decomposers that break down and recycle organic nutrients. None of the Archeognatha are considered pests.

Order: Zygentoma

Eg: Silverfish and firebrats

- Still known as Thysanura
- Many superficial similarities with Archeognatha
- Major difference is having a dicondylous mandible
- Ametabolous development



Sub-Class : Palaeoptera

- Lack the ability to fold their wings over the abdomen

Order: Ephemeroptera

Eg: Mayflies

- Meaning “short lived winged”
- Most primitive of winged insects
- Triangular forewings” hind wing smaller
- Wings held vertically above the body at rest
- Often aquatic
- Many nymph stages (living in water)
- Two flying stages (subimago and imago)
- Three long caudal filaments at the tip of the abdomen
- Hemimetabolous development



Order: Odonata

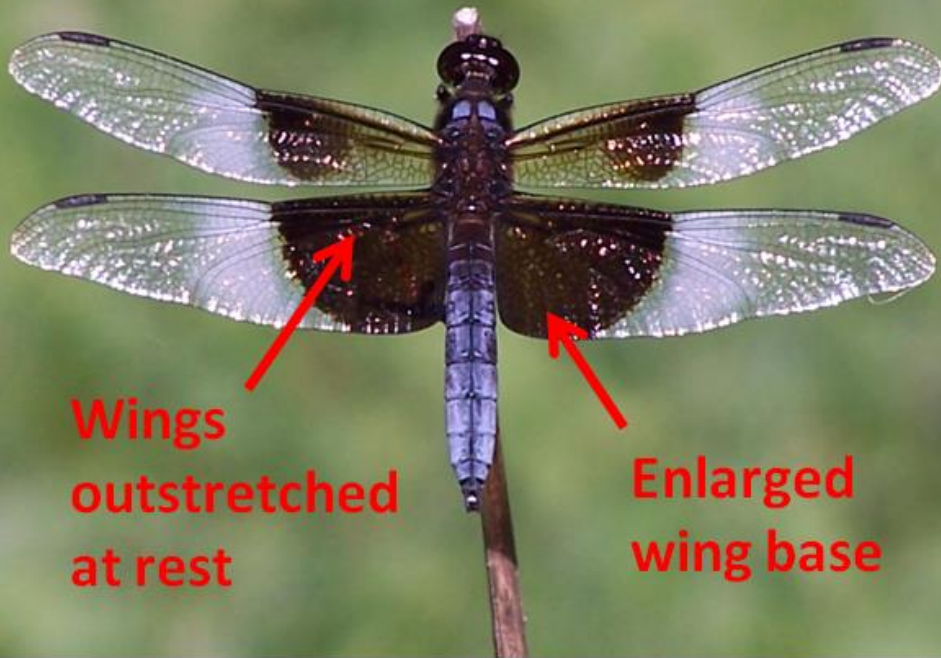
Eg: Dragonflies and damselflies

- “odonto-“, meaning tooth
- Minute antennae
- Extremely large eyes that fills most of the head
- Short, bristle-like antennae
- Two pairs of transparent membranous wings
- Wings have small veins
- Rectangular stigma (pigmented patch) near tip of each wing
- Long slender abdomen
- Aquatic larval stage (nymph) with tracheal gills and prehensil labium (extendible jaws)
- 25 families
- Hemimetabolous development

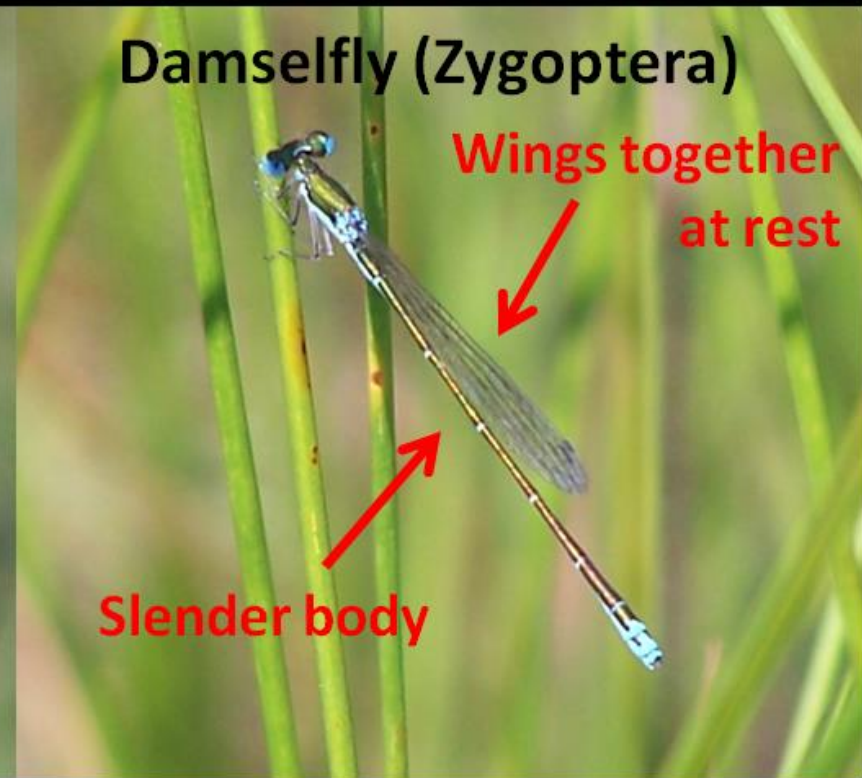


Dragonflies vs Damselflies

Dragonfly (Anisoptera)



Damselfly (Zygoptera)



Dragonflies (suborder Anisoptera)

Aeshnidae — Darners

Gomphidae — Clubtails

Libellulidae — Common skimmers

Damselflies (suborder Zygoptera)

Calopterygidae — Broadwinged damselflies

Coenagrionidae — Narrowwinged damselflies

Lestidae — Spreadwinged damselflies

Sub-Class : Polyneoptera

- Higher group of winged insects
- Have the ability to flex their wings so that they can be folded flat over the body
- Evolutionary advantage → protect the wings from damage while crawling through vegetation or other confined space
 - Lead to the development of thicker fore wings as protective covers for the hind wings and abdomen

Ex: Grasshoppers, crickets, earwigs, stick- insects, cockroaches,

Order: Mantophasmatodea

Eg: Heelwalkers

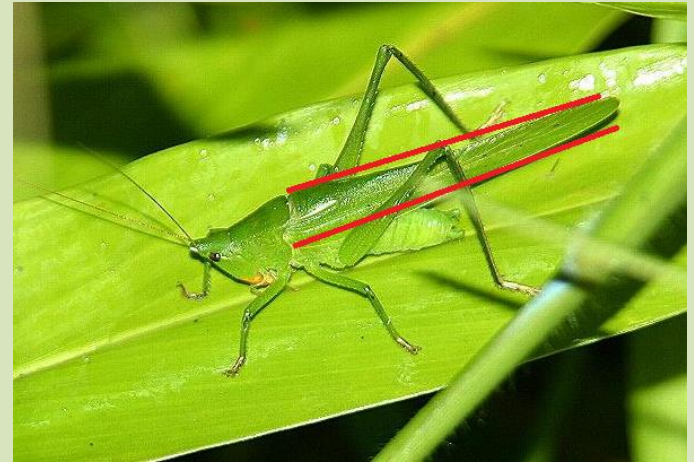
- Small predatory insects
- Mouthparts mandibulate, hypognathous
- 5-segmented tarsi
- Cerci short, 1-segmented
- Recently discovered (in 2001)
- Fewer than 20 species all known from Southern and East Africa
- They hold the distal end of their tarsi off the ground when walking → so called heelwalkers
- Phylogenetic relationship – not clear: closely related to Grylloblattodea
- Hemimetabolous development



Order: Orthoptera

Eg: Grasshopper

- Largest of Polyneopteran
- Means “straight wings”
- 2 pairs of wings
- Forewings – narrower than hind wing
- Wings harden or leathery at the base
- Held roof like overlapping the abdomen at rest
- Hind wing - membranous: held folded fan-like under the forewings at rest
- Mandibulate mouthparts
- Large compound eyes
- Antennae can vary from short to long
 - Short – grasshopper
 - Long - Katydid



- Hind leg enlarged and modified for jumping
- Incomplete metamorphosis/ hemimetabolous

Short horned grasshoppers

Gryllidae — House, field, and tree crickets

Gryllotalpidae — Mole crickets

Rhaphidophoridae — Cave and camel crickets

Long horned grasshoppers

Tettigoniidae — Katydid

Crickets

Gryllidae — House, field, and tree crickets

Gryllotalpidae — Mole crickets

Rhaphidophoridae — Cave and camel crickets



Order: Phasmida

Eg: stick insect/ leaf insects/ walkingstick

- Cryptic insect – shape and color similar to twigs or leaves
- Body shape variable – most are elongated and cylindrical or flattened and resemble sticks, leaves or grass
- 2 pairs of wings
- Some species are wingless
- Wings, if present, hardened forewings membranous hind wing
- Antennae are filiform – short or long
- Prothorax shorter than mesothorax or metathorax
- Nymphs resemble adults
- Hemimetabolous development

Order: Phasmida/ Phasmatodea

Eg: stick insect/ leaf insects/ walkingstick

- Cryptic insect – shape and color similar to twigs or leaves
- Body shape variable – most are elongated and cylindrical or flattened and resemble sticks, leaves or grass
- 2 pairs of wings
- Some species are wingless
- Wings, if present, hardened forewings membranous hind wing
- Antennae are filiform – short or long
- Mouthparts mandibulate, prognathous
- Prothorax shorter than meso- or metathorax
- Nymphs resemble adults. Hemimetabolous development



Order: Dictyoptera

Eg: Cockroaches, termites and mantids

- Cockroaches, termites and mantids have been considered separate orders previously
 - Cockroaches – Blattodea
 - Mantids – Mantodea
 - Termites – Isoptera
- Now these three groups are considered as sub-orders under order Dictyoptera

Order: Dictyoptera

Eg: Cockroaches, termites and mantids

- Cockroaches, termites and mantids have been considered separate orders previously
 - Cockroaches – Blattodea
 - Mantids – Mantodea
 - Termites – Isoptera
- Now these three groups are considered as sub-orders under order Dictyoptera
- Recent research has shown that Termites are actually a lineage of cockroaches and not a separate insect Order as previously thought

Roaches vs mantids

- have leathery forewings - tegmina
- wings are held like a roof over the abdomen
- have long and spiny middle and hind legs
- feet, or tarsi, nearly always have five segments
- chewing mouthparts
- long, segmented antennae
- Cerci are visible
- Hemimetabolous development



DKH GE



Roaches vs termites

- Genetical studies confirmed the relationship
- Comes from the same lineage
- Both insects seek shelter
- lay eggs in large batches
- some cockroaches and termites that lay eggs in individual cases/ specialized cases that enclose their eggs
- Include symbiotic microbes that certain termites and wood roaches had in common
- physical similarities between the young
- Some have slender abdomen and larger head
- In both groups there are some that fly while most walk on the floor or walls to reach their destinations
- Both tend to make tunnels in wood

Sub Order: Mantodea

Eg: mantids

- Head triangular with well-developed compound eyes
- Elongated prothorax
- Grasping (raptorial) front legs for catching prey
- Filiform antennae
- Mouthparts mandibulate, hypognathous
- Front wings thickened, more slender than hind wings
- Tarsi 5-segmented
- Cerci short, multi-segmented
- Immature structurally similar to adults

Sub Order: Blattodea

Eg: cockroaches

- Body elongate oval in shape
- Pronotum forms transverse oval just behind head
- Short, multi-segmented cerci
- Long. filiform antennae
- Legs adapted for running
- tarsi 5-segmented
- Front wings thickened
- hind wings membranous, pleated
- Cerci short, multi-segmented



Sub Order: Isoptera

Eg: termites

- Body pale in color,
- somewhat ant-like in appearance
- Wide junction between thorax and abdomen
- Workers are cream-colored with beaded antenna and oval heads
- Soldiers have larger heads with prominent mandible
- Compound eyes small or absent
- Head large and cylindrical or small and round
- Antennae beaded
- Mouthparts chewing; sometimes with large mandibles



Order: Dermaptera

Eg: earwigs

- The terminal, forceps-like cerci
- Antennae slender, beaded
- Mouthparts mandibulate, prognathous
- Tarsi 3-segmented
- Front wings short and leathery
- Hind wings semicircular and pleated
- Cerci enlarged to form pincers (forceps)
- Immature structurally similar to adults
- Hemimetabolous development



Order: Grylloblattaria

Eg: mole cricket, rock crawlers

- Small and obscure order
- Antennae slender, filiform
- Mouthparts mandibulate, hypognathous
- Body cylindrical
- Tarsi 5-segmented
- Secondarily wingless
- Cerci long, 8-segmented
- Hemimetabolous development



Sub-Class : Paraneoptera

- Paraneoptera consists of 4 insect orders, the bark lice, true lice, thrips and hemipterans
- The evolutionary history of the Paraneoptera is a history beautifully reflected in structure and function of their mouthparts.

Order: Hemiptera

Sub-order: Heteroptera

Eg: true bugs

- the elongated rostrum is unique to the Hemiptera

- Antennae slender with 4-5 segments
- Proboscis 3-4 segmented, arising from front of head and curving below body when not in use
- Pronotum usually large, trapezoidal or rounded
- Triangular scutellum present behind pronotum



- Front wings with basal half leathery and apical half membranous (hemelytra). Wings lie flat on the back at rest, forming an “X”
- Tarsi 2- or 3-segmented



Common Families:

Belostomatidae — Predaceous Diving Bugs

- Front legs raptorial and bearing a single claw
- Body flattened and oval in shape



Common Name: Diving bug
Class: Insecta
Order: Hemiptera **Family:** Belostomatidae
Ecological niche: predator
Distinguishing characteristics: oval body (often large) with front legs adapted for catching prey (raptorial); males of some species carry eggs on their back



Common Families:

Coreidae — Squash bugs and Leaffooted Bugs

- Conspicuous scent glands on sides of thorax between middle and hind coxae
- Head much narrower than thorax
- Ocelli present
- Many parallel veins in membranes of front wings
- In leaffooted bugs, the hind tibia are expanded into thin, leaf-like plates



Common Families:

Corixidae — Water Boatmen

- Front tarsi 1-segmented and scoop-shaped
- Hind legs lack claws
- Tarsi of hind legs bear a fringe of long hairs as an adaptation for swimming



Common Families:

Gerridae — Water Striders

- Long, slender legs
- Tarsal claws anteapical (located before tip of leg)



Common Families:

Miridae — Plant Bugs

- Cuneus present on front wings
- Ocelli absent
- Wings angle down behind abdomen



Common Families:

Notonectidae — Backswimmers

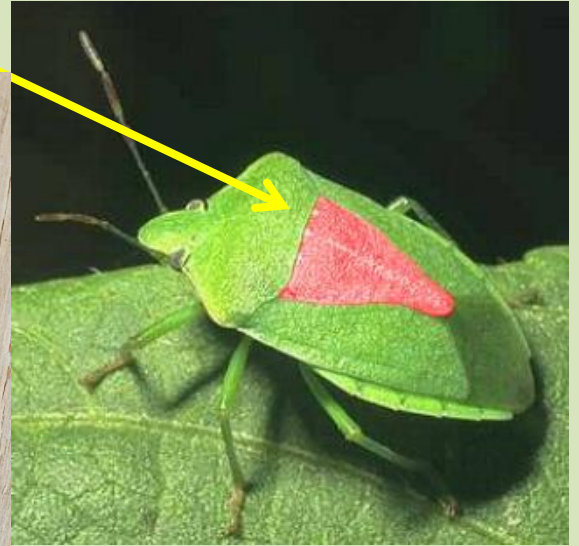
- Keel-shaped dorsal side is silvery-white
- Tibia and tarsi of hind legs bear a fringe of long hairs as an adaptation for swimming
- Hind legs lack claws
- The dorsal side of the backswimmer is convex and darker than the ventral side.
- These insects are predaceous and pursue their prey by sneaking under the victim's body and sucking on their body fluids.



Common Families:

Pentatomidae — Stink Bugs

- Five-segmented antennae
- Three-segmented tarsi and tibiae lack hairs
- Large triangular scutellum in center of back
- Shield-shaped body



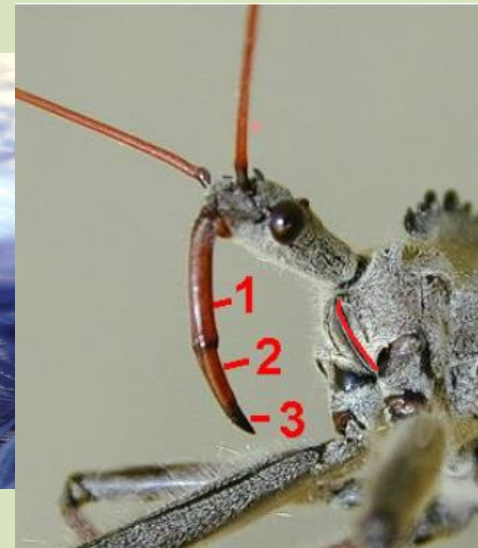
Common Families:

Reduviidae — Assassin Bugs

- Three-segmented proboscis
- Prosternal groove between front legs
- Head elongate, narrowing behind eyes
- Lateral margins of abdomen extend beyond wings
- Raptorial front legs are covered with hairs and spines
- abdomen is often wider than the rest of the body



DKH GE



Common Families:

Coreidae — Squash bugs and Leaffooted Bugs

- Conspicuous scent glands on sides of thorax
between middle and hind coxae
- Head much narrower than thorax
- Ocelli present
- Many parallel veins in membranes of front wings
- In leaffooted bugs, the hind tibia are expanded into
thin, leaf-like plates

Corixidae — Water Boatmen

Gerridae — Water Striders

Miridae — Plant Bugs

Notonectidae — Backswimmers

Pentatomidae — Stink Bugs

Reduviidae — Assassin Bugs

Order: Hemiptera

Sub-order: Homoptera

Eg: leafhoppers, planthoppers, treehoppers, cicads, aphids, whiteflies, scale insects, phyllids

- Uniform texture of the wings
- Short proboscis (beak) emerges near back of head
- Wings held tent-like over abdomen
- Many with bristle-like antennae
- Many with wedge shaped head
- Hemimetabolous development
- Immature structurally similar to adults

Common Families:

Aphidoidea — Aphids

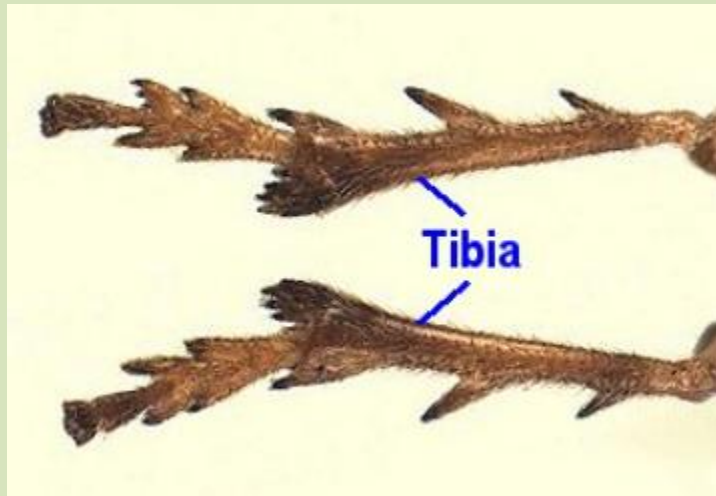
- Cornicles usually present on abdomen
- Small, soft-bodied, with visible legs and antennae
- Adults may be winged or wingless



Common Families:

Cercopidae — Spittlebugs

- Hind tibia have a “crown” of large spines at the distal end
- Short, bristle-like antennae



Common Families:

Cicadellidae — Leafhoppers

- Comb-like rows of large spines along tibia of hind legs
- Short, bristle-like antennae



Common Families:

Cicadidae — Cicadas

- Three ocelli form a triangle on top of head
- Short, bristle-like antennae
- Large body size
- Membranous wings
- range in color
- male cicadas sing in a high pitch, pulsating tone
- lay their eggs in slit twigs of trees



Common Families:

Coccoidea — Scale Insects

- Sessile insects that secrete a protective cover over their soft bodies
- Legs and antennae not apparent



Common Families:

Membracidae — Treehoppers

- Large pronotum covers the head and extends backwards over the abdomen
- pronotum is armed with spines and needles to use as a defense mechanism
- Some species of tree hoppers have a distinctive 'humpbacked' appearance
- Short, bristle-like antennae



Order: Thysanoptera

Eg: Thrips

- refers to the slender wings that bear a dense fringe of long hairs
- Tiny, tapered body (spindle-shaped – widest in middle)
- Body cylindrical or spindle-shaped
- Head narrow anteriorly forming a conical mouth opening
- Antennae short, 6-10 segments
- Front and hind wings slender, rod-like. Fringe of long hairs on each slender wing
- Many species are secondarily wingless
- Mouthparts concealed within head capsule
- Tarsi 1-2 segmented, with eversible adhesive bladders apically

- Hemimetabolous development
- closely related to Hemiptera and Psocodea
- Thysanoptera is divided into the two suborders Terebrantia and Tubulifera
- suborders can be distinguished by the shape of the last abdominal segment:
 - short and pointed in the Terebrantia,
 - long and tubular in the Tubulifera.

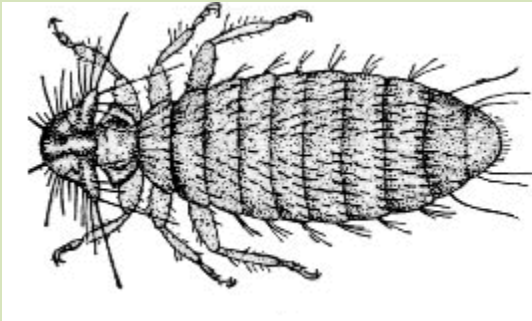


Order: Phtiiraptera

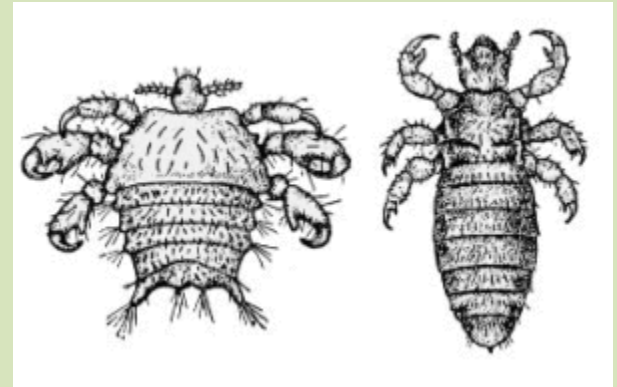
Eg: sucking and biting lice

- This was grouped under Psocodea. But now Psocodea is taken as a wider group
- This has been divided to Phthiraptera and Psocoptera
- four suborders
 1. Anoplura – sucking lice
 2. Amblycera – active chewing and biting lice on mammals
 3. Ischnocera – lethargic chewing and biting lice on mammals

- All adult lice are wingless
- dorsoventrally flattened
- eyes are reduced or absent
- usually have well-developed claws for clinging to their host
- Most lice are just a few millimetres long
- 'chewing' suborders the mandibulate mouthparts are used to bite and chew
- more specialised Anoplura the mouthparts are highly modified into piercing stylets
- Sucking lice have a conical head that is narrower than the thorax
- Chewing lice have a broad, flat head that is as wide as, or wider than the thorax
- Hemimetabolous development



Chewing lice



Sucking lice



Sub-Class : Endopterygota

- The larvae look very different to adults
- Undergo metamorphosis in a pupa
- Wings are developed
- Holometabolous development

Order: Coleoptera

Eg: beetles

- refers to the modified front wings which serve as protective covers for the membranous hind wings
- Hardened front wings (elytra) meet in a straight line down the center of the back
- Mandibulate mouthparts. Chewing mouthparts (sometimes located at the tip of a beak or snout)

- Hind wings large, membranous, folded beneath the elytra
- Tarsi 2- to 5-segmented
- Immature: head well developed with ocelli and chewing mouthparts
 - 3 pairs of thoracic legs
 - No abdominal prolegs
 - Campodeiform, scarabaeiform and elateiform is present

Common Families:

Cantharidae — Soldier Beetles

Carabidae — Ground Beetles

Cerambycidae — Longhorned Beetles

Chrysomelidae — Leaf Beetles

Coccinellidae — Lady Beetles

Curculionidae — Weevils

Dytiscidae — Predaceous Diving Beetles

Elateridae — Click Beetles

Gyrinidae — Whirligig Beetles

Hydrophilidae — Water Scavenger Beetles

Meloidae — Blister Beetles

Passalidae — Bess Beetles

Scarabaeidae — Scarab Beetles

Silphidae — Carrion Beetles

Staphylinidae — Rove Beetles

Tenebrionidae — Darkling Beetles

Common Families:

Carabidae — Ground Beetles

- distinct from one another in their size, shape, and color
- night feeders and the majority of them are predaceous
- they can be found under stones, logs, or piles of debris
- filiform antennae
- large hind trochanters
- elytra with longitudinal rows of grooves



Common Families:

Chrysomelidae — Leaf Beetles

- smaller than other families of Coleoptera
- do not get bigger than 12 mm in length
- antennae are much shorter than the body itself
- most noticeable characteristic on this beetle is the 4 segmented tarsi
- the 3rd tarsal segments notched where they connect with the 4th



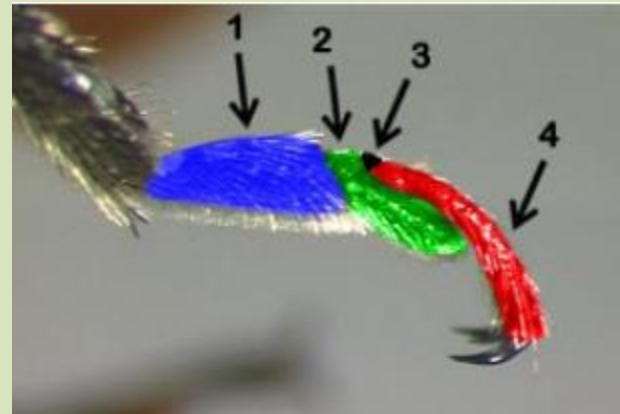
Common Families:

Coccinellidae — Lady Beetles

- adults are oval and brightly colored
- Black with red spots or red/orange with black spots
- appear to have three tarsal segments but actually have four
- small bead-like third segment is nested within the second segment
- Larvae are flattened and covered with spines
- Both the larvae and adults are predaceous on aphids



DKH GE





Common Families:

Curculionidae — Weevils

- defining characteristic of the weevils is the proboscis
- short and wide or long and slender
- chewing mouthparts are reduced in size
- can be seen at the very end of the proboscis
- The head is prolonged and leans forward to fit into the snout
- the antennae protrude from the sides of the snout
- Antennae elbowed
- 5-segmented tarsi
- beetles will sing by rubbing their elytra against their abdomen
- The scarabaeiform larvae can be seen burrowing in twigs, and large nuts





Common Families:

Scarabaeidae — Scarab Beetles

- Scarab beetles are abundant
- diverse in size but can be recognized by their oval bodies,
- five segmented tarsi,
- Spiny legs
- lamellate antennae,
- a front tibiae that is scalloped.
- usually found feeding on dung, plants, or flowers



Order: Diptera

Eg: true flies

- true flies have only a single pair of wings
- Front wings membranous; hind wings modified as small halteres
- Haustellate/ sponging mouthparts
- Mesothorax larger than pro- or metathorax
- Tarsi 5-segmented



Common Families:

Asilidae — Robber Flies

Calliphoridae — Blow Flies

Chironomidae — Midges

Culicidae — Mosquitoes

Syrphidae — Flower Flies

Tabanidae — Horse Flies and Deer Flies

Tachinidae — Tachinid Flies

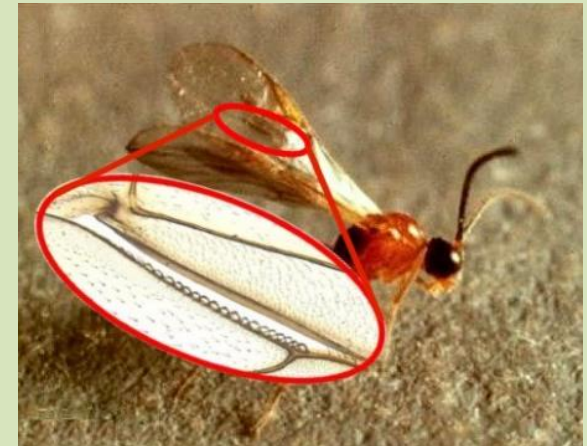
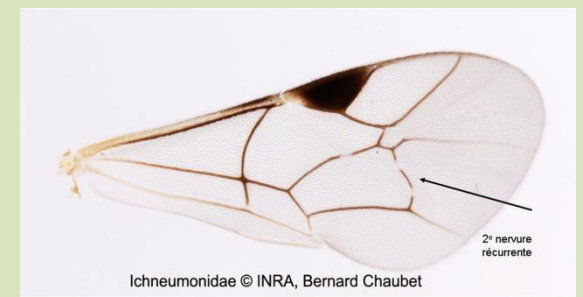
Tipulidae — Crane Flies



Order: Hymenoptera

Eg: wasps, ant, bees and sawflies

- Wasp-waist present in ants, bees, and wasps
- Mandibulate mouthparts
- Triangular stigma in front wing of sawflies, horntails, and some wasps
- Hind wings smaller than front wings
- Hamuli (tiny hooks on costa of hind wing) hold front and hind wings together
- Chewing mouthparts – except in bees
- Compound eyes well developed.
- Tarsi usually 5-segmented



Common Families:

Apidae — Bumble Bees and Honey Bees

Formicidae — Ants

Halictidae — Sweat Bees

Ichneumonidae — Ichneumon Wasps

Sphecidae — Sand Wasps, Digger Wasps, and Mud-daubers

Scoliidae — Scoliid Wasps

Vespidae — Yellowjackets, Hornets, Paper Wasps, and Potter Wasps

Common Families:

Apidae — Bumble Bees and Honey Bees

- First segment of hind tarsi enlarged and flattened; usually bearing a “pollen basket”
- Their maxillae and labium have evolved to form a tongue for collecting nectar
- Body heavily clothed with branched hairs



Common Families:

Formicidae — Ants

- Peduncle located between thorax and abdomen — first (or first and second) abdominal segments separate from rest of abdomen
- Elbowed antennae
- Wingless workers; winged swarmers (reproductives)
- Colonies of ants vary in size and operate within a three-caste system



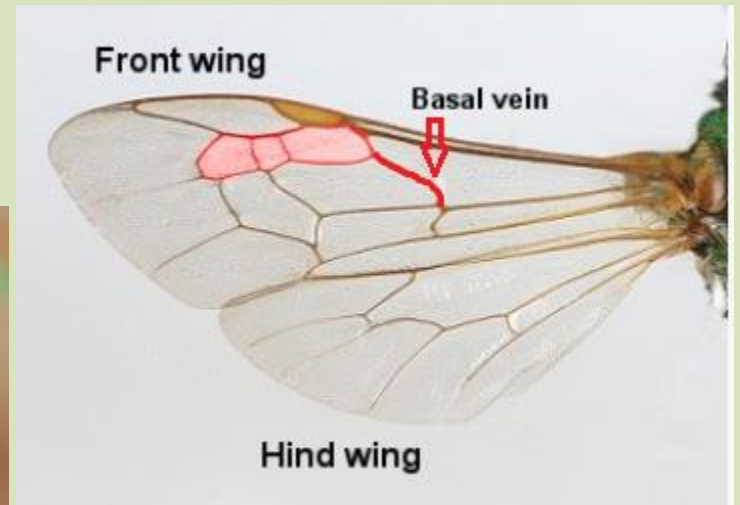
DKH GE



Common Families:

Halictidae — Sweat Bees

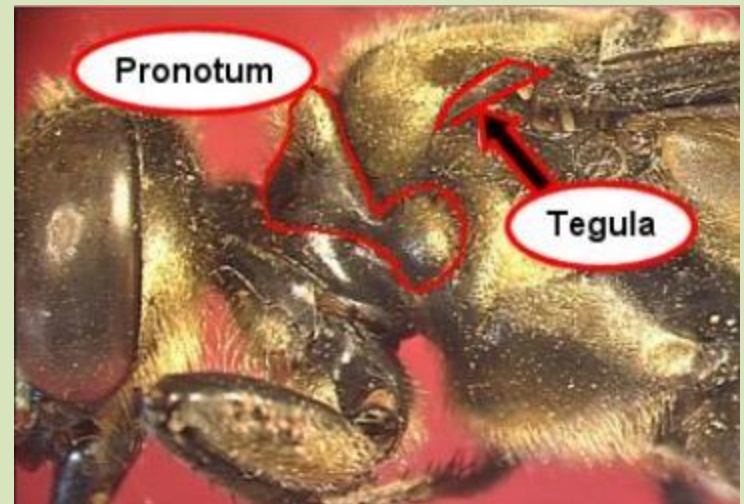
- can be different colors but most often stand out when they are metallic
- curved basal vein
- and two or three submarginal cells in the forewing
- solitary in nature
- nest in the ground
- feed on pollen



Common Families:

Sphecidae — Sand Wasps, Digger Wasps, and Mud-daubers

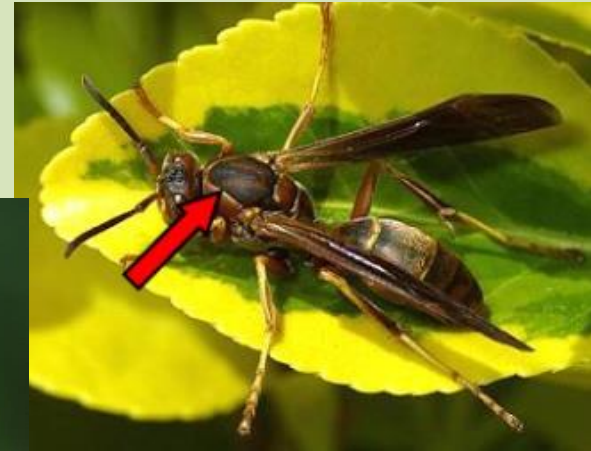
- Lateral margin of pronotum forms a distinct lobe that does not touch the tegula
- Body relatively devoid of hairs
- Basal segment of tarsi not enlarged or flattened
- The larvae are parasitic while the adults are entomophagous in nature



Common Families:

Vespidae — Yellowjackets, Hornets, Paper Wasps, and Potter Wasps

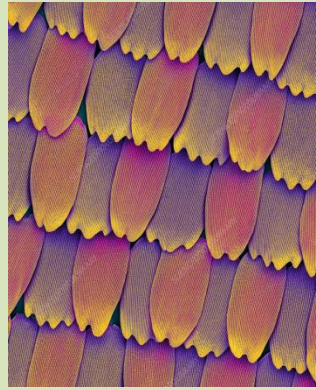
- Posterior margin of pronotum distinctly “U”-shaped
- Forewings fold in half longitudinally
- Many species of Vespidae are eusocial, living in a colony comprised of three castes: queens, female workers, and drones
- Only the workers are able to sting



Order: Lepidoptera

Eg: Butterflies and moths

- “lepidō” for scale and “ptera” for wings
- Mouthparts form a coiled tube (proboscis) beneath the head
- Antennal type:
Butterflies: knobbed or hooked at tip
Moths: thread-like, spindle-shaped, or comb-like
- Front wings large, triangular; hind wings large, fan-shaped
- Body and wings covered with small, overlapping scales



Common Families:

Butterflies

Danaidae — Milkweed Butterflies

Hesperiidae — Skippers

Nymphalidae — Brush-footed Butterflies

Papilionidae — Swallowtail Butterflies

Pieridae — Whites and Sulfurs

Moths

Arctiidae — Tiger Moths

Noctuidae — Underwing Moths

Saturniidae — Silk Moths

Sphingidae — Hawk Moths

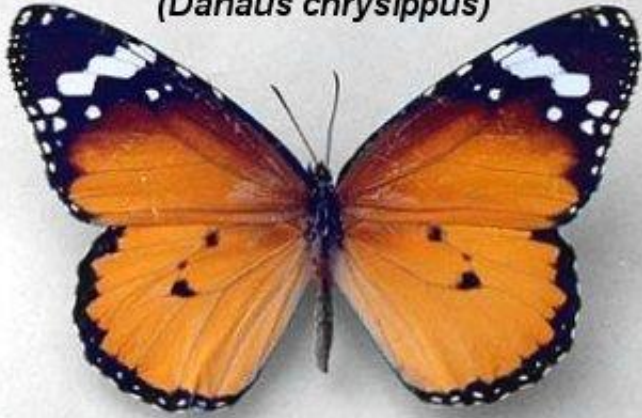
Common Families:

Butterflies

Danaidae — Milkweed Butterflies, plain tiger, common tiger, monarch butterfly

- Orange color with black markings along wing veins - especially on ventral side
- short third anal vein in the front wing
- Many of the species are bright orange with black markings along the veins of the wings.
- Many of the larvae feed on milkweed

The Traveler
(*Danaus chrysippus*)



Common Families:

Butterflies

Hesperiidae — Skippers, common
banded awl



- small to medium-size butterflies
- orange or brown wings and stout bodies
- antennae are distinctively hooked or recurved at the tip and widely separated at the base
- The head is relatively broad with widely separated eyes and antennae
- Many species rest with their front and hind wings held at different angles



Common banded awl



Water snow flat



Common Families:

Butterflies

Nymphalidae — Brush-footed Butterflies, common sailor, chocolate soldier, common crow

- Front legs reduced in size and lacking claws; usually held against the body and not used for walking





common sailor



common crow



chocolate soldier



Sri Lanka tree
brown

Common Families:

Butterflies

Papilionidae — Swallowtail Butterflies, lime butterfly, common rose, banded peacock, crimson rose

- relatively large in size
- Front legs well-developed
- Stout spur (epiphysis) on front tibia
- Characteristic tail on hind wings (swallowtails)
- Red or orange spot on hind wing (parnassians) found mostly at high elevations





Crimson rose



Banded peacock



Lime butterfly



DKH GE Blue Bottle

Common Families:

Butterflies

Pieridae — Whites and Sulfurs, Jezebel, Sri Lankan Lesser Albatross, Lemon emigrant

- medium to small sized and are distributed worldwide
- the front legs are well-developed with bifid tarsal claws
- The pupae are often elongate chrysalids



Jezebel



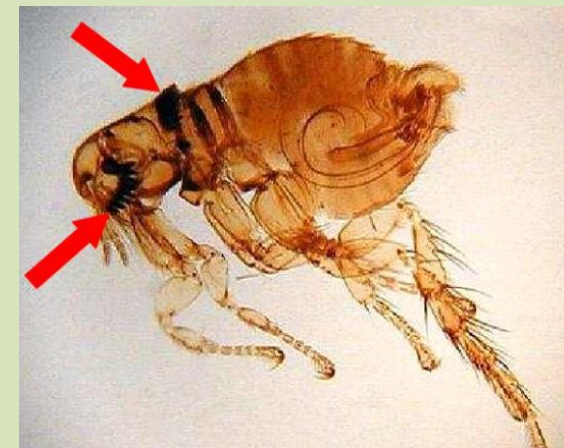
DKH GE

Lemon emigrant

Order: Siphonaptera

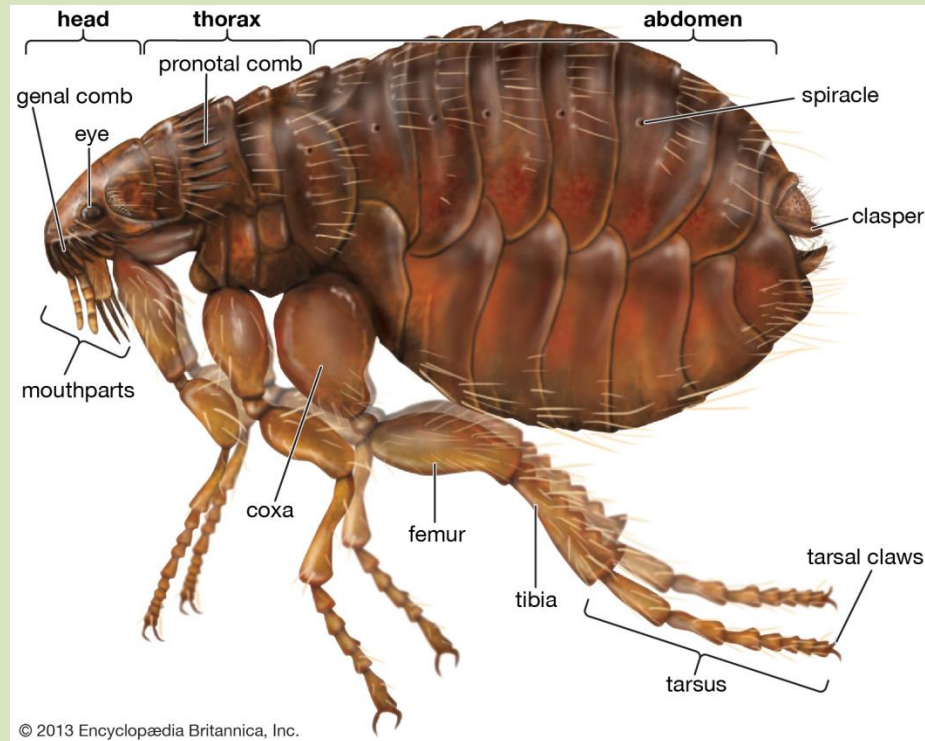
Eg: fleas

- secondarily wingless insects whose mouthparts are adapted for piercing skin and sucking blood
- Body bilaterally flattened
- Mouthparts piercing and sucking/suctorial (haustellate)
- Large bristles (ctenidia) often present on head or thorax (genal and pronotal combs)
- Hind femur enlarged, adapted for jumping
- Larval form is maggot-like/vermiform
- Holometabolous development



Major family

Pulicidae (common fleas) — This family includes most species with economic or medical importance: the cat flea (*Ctenocephalides felis*), the dog flea (*C. canis*), the human flea (*Pulex irritans*), the rabbit flea (*Spilopsyllus cuniculi*), and the oriental rat flea (*Xenopsylla cheopis*).



THANK YOU!

